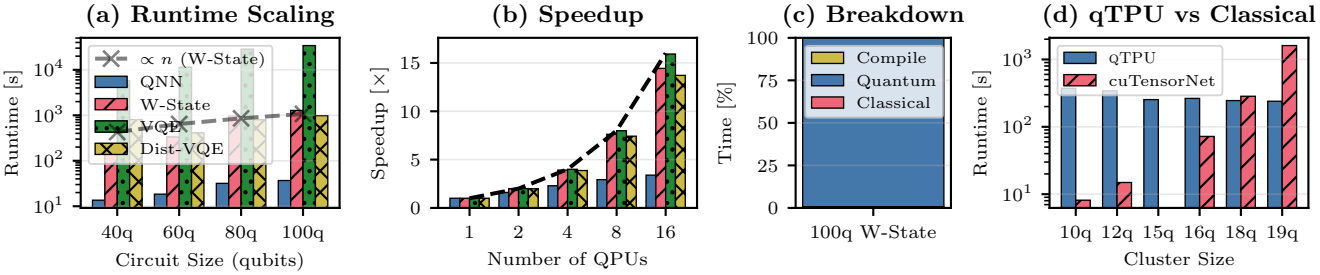
(a) Runtime Scaling
Runtime [s]
10^4
10^3
10^2
10^1
$\propto n$ (W-State)
QNN
W-State
VQE
Dist-VQE
40q
60q
80q
100q
Circuit Size (qubits)
(b) Speedup
Speedup [×]
15
10
5
0
1
2
4
8
16
Number of QPUs
(c) Breakdown
Time [%]
100
75
50
25
0
Compile
Quantum
Classical
100q W-State
(d) qTPU vs Classical
Runtime [s]
10^3
10^2
10^1
qTPU
cuTensorNet
10q
12q
15q
16q
18q
19q
Cluster Size